

Mid-Term Exam for 《Calculus》

2022-2023 2nd Semester

Class: _____ Name: _____ Sit NO.: _____ Total Score: _____

Note: 1. Please clearly write down the private information needed before exam.

2. Please clearly write down the answers on the paper or answer sheet.

3. Full mark: 100

4. Duration of Exam: 120 minutes

1. Please fill the correct answers in the following blanks. (18 marks)

(1) If $\vec{a} = \langle -1, 2, 2 \rangle$, $\vec{b} = \langle 2, -1, 2 \rangle$, $\vec{c} = \langle 1, 0, 0 \rangle$, then $\vec{a} \times \vec{b} \cdot \vec{c}$ is

$\langle 6, 6, -3 \rangle \cdot \langle 1, 0, 0 \rangle = 6$ _____;

(2) Equation $xyz + \sqrt{x^2 + y^2 + z^2} = \sqrt{2}$ defines an implicit function $z = f(x, y)$, the differential of the dependent variable dz at point $(1, 0, -1)$ is $dx - \sqrt{2}dy$ _____;

(3) Evaluate iterated integrals:

$$\int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx = \frac{3}{8}e - \frac{1}{2}\sqrt{e}$$
 _____;

(4) The equation of the tangent line to the curve $\begin{cases} z = x^2 + y^2 \\ x^2 + 2y^2 + 3z^2 = 8 \end{cases}$ at point $P(0, \frac{2\sqrt{3}}{3}, \frac{4}{3})$ is

$$\left\{ \begin{array}{l} y = \frac{2\sqrt{3}}{3} \\ z = \frac{4}{3} \end{array} \right.$$
 _____;

(5) The radius of convergence of $\sum_{n=1}^{\infty} \frac{n}{2^n} x^{2n}$ is $\sqrt{2}$ _____;

(6) Suppose $f(x, y)$ attains its local extreme value at (x_0, y_0) . If $f(x, y)$ is differentiable at

(x_0, y_0) , then the directional derivative of f at (x_0, y_0) in the arbitrary direction is _____0_____.

2. (6 marks) Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2}$ does not exist.

Solution:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=kx (k \neq 1)}} \frac{k^2 x^4}{k^2 x^4 + (1-k)^2 x^2} = \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=kx (k \neq 1)}} \frac{k^2 x^2}{k^2 x^2 + (1-k)^2} = 0,$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=x}} \frac{x^4}{x^4} = 1$$

So the limit does not exist.

3. (10 marks) Let $f(x, y) = \begin{cases} (x+y)^2 \sin \frac{1}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$

(1) Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$;

(2) Determine the continuity of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the origin;

(3) Determine the differentiability of $f(x, y)$ at the origin. Please show the reason.

Solution:

$$(1) f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^2 \sin \frac{1}{(\Delta x)^2}}{\Delta x} = 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{(\Delta y)^2 \sin \frac{1}{(\Delta y)^2}}{\Delta y} = 0$$

$$f_x(x, y) = \begin{cases} 2(x+y) \sin \frac{1}{x^2 + y^2} + (x+y)^2 \cos \frac{1}{x^2 + y^2} \frac{-2x}{(x^2 + y^2)^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$f_y(x, y) = \begin{cases} 2(x+y) \sin \frac{1}{x^2 + y^2} + (x+y)^2 \cos \frac{1}{x^2 + y^2} \frac{-2y}{(x^2 + y^2)^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$$

(2) $\lim_{(x,y) \rightarrow (0,0)} f_x(x,y) = \lim_{(x,y) \rightarrow (0,0)} [2(x+y) \sin \frac{1}{x^2+y^2} + (x+y)^2 \cos \frac{1}{x^2+y^2} \frac{-2x}{(x^2+y^2)^2}]$ doesn't exist.

$\frac{\partial f}{\partial x}$ is not continuous at the origin, the same to the $\frac{\partial f}{\partial y}$.

(3)

$$\lim_{(\Delta x, \Delta y) \rightarrow (0,0)} \frac{f(0+\Delta x, 0+\Delta y) - f(0,0) - [f_x(0,0)\Delta x + f_y(0,0)\Delta y]}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} =$$

$$\lim_{(\Delta x, \Delta y) \rightarrow (0,0)} \frac{(\Delta x + \Delta y)^2 \sin \frac{1}{(\Delta x)^2 + (\Delta y)^2}}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = 0,$$

$f(x,y)$ is differentiable at the origin.

4.(6 marks) Let $\xi = \frac{y}{x}$, $\eta = y$, the second partial derivatives of u are continuous. Try to prove

the equation $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 0$ can be transformed into the equation $\eta^2 u_{\eta\eta} = 0$.

Solution:

$$u_x = u_\xi \left(-\frac{y}{x^2}\right), \quad u_{xx} = u_{\xi\xi} \frac{y^2}{x^4} + u_\xi \frac{2y}{x^3}$$

$$u_{xy} = -\frac{u_\xi}{x^2} + \left(u_{\xi\xi} \frac{1}{x} + u_{\xi\eta}\right) \left(-\frac{y}{x^2}\right)$$

$$u_y = u_\xi \frac{1}{x} + u_\eta, \quad u_{yy} = \frac{1}{x^2} u_{\xi\xi} + \frac{2}{x} u_{\xi\eta} + u_{\eta\eta}$$

5.(6 marks) Find the gradient of function $z = x^2 - xy + y^2$ at point $(-1,1)$. In what direction

does $z = x^2 - xy + y^2$ decrease most rapidly? And in what direction does $z = x^2 - xy + y^2$ stay at the same level?

Solution:

(1) $\text{grad } z|_{(-1,1)} = \langle 2x - y, -x + 2y \rangle|_{(-1,1)} = \langle -3, 3 \rangle;$

(2) $z = x^2 - xy + y^2$ decreases most rapidly in the direction $\langle 1, -1 \rangle;$

$z = x^2 - xy + y^2$ stays at the same level in the direction $\langle 1, 1 \rangle, \langle -1, -1 \rangle.$

6.(6 marks) Find the equations of the tangent plane and the normal line to the given surface

$$e^{\frac{x}{z}} + e^{\frac{y}{z}} = 3 \text{ at the indicated point } (\ln 2, 0, 1).$$

Solution:

$$F(x, y, z) = e^{\frac{x}{z}} + e^{\frac{y}{z}} - 3, F_x = \frac{1}{z} e^{\frac{x}{z}}, F_y = \frac{1}{z} e^{\frac{y}{z}}, F_z = -\frac{1}{z^2} (x e^{\frac{x}{z}} + y e^{\frac{y}{z}})$$

$$\text{normal vector at } (\ln 2, 0, 1) \text{ is } \left\langle \frac{1}{z} e^{\frac{x}{z}}, \frac{1}{z} e^{\frac{y}{z}}, -\frac{1}{z^2} (x e^{\frac{x}{z}} + y e^{\frac{y}{z}}) \right\rangle \Big|_{(\ln 2, 0, 1)} = \langle 2, 1, -2 \ln 2 \rangle$$

$$\text{tangent plane: } 2(x - \ln 2) + (y - 0) - 2 \ln 2(z - 1) = 0$$

$$\text{normal line: } \frac{x - \ln 2}{2} = \frac{y}{1} = \frac{z - 1}{-2 \ln 2}$$

7. (6 marks) Find the minimum distance between the origin and the surface $x^2 y - z^2 + 9 = 0$.

Solution:

Minimize the square of the distance $f(x, y, z) = x^2 + y^2 + z^2$, subject to

$$g(x, y, z) = x^2 y - z^2 + 9 = 0$$

Let

$$L(x, y, z) = x^2 + y^2 + z^2 - \lambda(x^2 y - z^2 + 9)$$

$$\begin{cases} L_x = 2x - 2\lambda xy = 0 \\ L_y = 2y - \lambda x^2 = 0 \\ L_z = 2z + 2\lambda z = 0 \\ L_\lambda = x^2 y - z^2 + 9 = 0 \end{cases}$$

Solve these equations, we get eight critical points :

$$(0, 0, \pm 3), (\pm 3\sqrt[6]{2/9}, -\sqrt[3]{9/2}, 0), (\pm\sqrt{2}, -1, \pm\sqrt{7})$$

$f((\pm 3\sqrt[6]{2/9}, -\sqrt[3]{9/2}, 0)) = 3 \cdot 2^{\frac{2}{3}} \cdot 9^{\frac{2}{3}}$ is the minimum distance of the origin and the surface.

8.(6 marks) Test for the convergence or divergence of $\sum_{n=1}^{\infty} \frac{n}{n5^n + 2}$.

Solution:

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n+1}{(n+1)5^{n+1} + 2} \cdot \frac{n5^n + 2}{n} = \frac{1}{5} < 1$$

$$\sum_{n=1}^{\infty} \frac{n}{n5^n + 2} \text{ is convergent.}$$

9.(6 marks) Find the convergence set and the sum function of the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x+2)^{2n}}{n \cdot 3^{n+1}}.$$

Solution : The radius of convergence $R = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{\frac{1}{n3^{n+1}}}{\frac{1}{(n+1)3^{n+2}}}} = \sqrt{3}$

$$x+2 = \pm\sqrt{3}, \sum_{n=1}^{\infty} (-1)^n \frac{3^n}{n3^{n+1}} = \frac{1}{3} \sum_{n=1}^{\infty} (-1)^n \frac{1}{n} \text{ converges.}$$

So the convergence set is $[-2-\sqrt{3}, -2+\sqrt{3}]$.

$$\text{Let } S(x) = \sum_{n=1}^{\infty} (-1)^n \frac{(x+2)^{2n}}{n3^{n+1}}, x \in [-2-\sqrt{3}, -2+\sqrt{3}], t = (x+2)^2$$

$$A(t) = \sum_{n=1}^{\infty} (-1)^n \frac{t^n}{n3^{n+1}} = \sum_{n=1}^{\infty} (-1)^n \frac{1}{3n} \left(\frac{t}{3}\right)^n,$$

$$A'(t) = \sum_{n=1}^{\infty} (-1)^n \frac{1}{9} \left(\frac{t}{3}\right)^{n-1} = \frac{-1}{9+3t},$$

$$A(t) = -\frac{1}{3} [\ln(9+3t) - \ln 9],$$

$$S(x) = -\frac{1}{3} [\ln(9+3(x+2)^2) - \ln 9].$$

10.(6 marks) Expand the function $f(x) = \ln \frac{1+x}{1-x}$ into power series of x .

Solution :

$$f(x) = \ln(x+1) - \ln(1-x) \quad -1 < x < 1$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$-(-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \dots)$$

$$= 2 \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1} \quad (-1 < x < 1)$$

11.(6 marks) Let $f(x) = \begin{cases} e^{-1/x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$, find the Maclaurin series for $f(x)$. Does the

Maclaurin series represent $f(x)$? Why?

$$\text{Solution : } f^{(n)}(0) = 0, (n = 0, 1, 2, \dots)$$

$$f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 0 = S(x)$$

$$S(x) \neq f(x) \text{ except } x = 0.$$

12.(6 marks) Find $\iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy$, D is bounded by $x+y=1$, $x=0$ and $y=0$.

Solution :

$$\text{let } \begin{cases} u = x+y \\ v = y \end{cases} \Rightarrow \begin{cases} x = u-v \\ y = v \end{cases}$$

$$D \rightarrow D'$$

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1$$

$$\iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy = \iint_{D'} \frac{v}{u} e^{u^2} du dv, D' \text{ is bounded by } v=u, v=0 \text{ and } u=1$$

$$= \int_0^1 du \int_0^u \frac{v}{u} e^{u^2} dv = \int_0^1 \frac{u}{2} e^{u^2} du = \frac{1}{4} (e-1).$$

13.(6 marks) Find the volume of the solid bounded by the surface $z=xy$ and planes $z=x+y$, $x+y=1$, $x=0$, $y=0$.

Solution :

$$V = \iiint_{\Omega} 1 dV = \iint_{D_{xy}} dx dy \int_{xy}^{x+y} 1 dz = \int_0^1 dx \int_0^{1-x} (x+y-xy) dy = \frac{7}{24}$$

14.(6 marks) Calculate $\iiint_{\Omega} (x^2 + y^2) dV$, Ω is bounded by the planes $z=2$, $z=8$ and the surface that results when the parabola $y^2=2z$ in the yz -plane is revolved about the z -axis.

Solution :

$$\text{Revolution surface : } x^2 + y^2 = 2z$$

$$\Omega_1 : 0 \leq \theta \leq 2\pi, 0 \leq r \leq 4, \frac{r^2}{2} \leq z \leq 8$$

$$\Omega_2 : 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, \frac{r^2}{2} \leq z \leq 2$$

$$\iiint_{\Omega} (x^2 + y^2) dV = \iiint_{\Omega_1} (x^2 + y^2) dV - \iiint_{\Omega_2} (x^2 + y^2) dV$$

$$= \int_0^{2\pi} d\theta \int_0^4 dr \int_{\frac{r^2}{2}}^8 r^2 \cdot r dz - \int_0^{2\pi} d\theta \int_0^2 dr \int_{\frac{r^2}{2}}^2 r^2 \cdot r dz$$

$$= \frac{4^5}{3} \pi - \frac{2^5}{6} \pi = 336\pi$$